

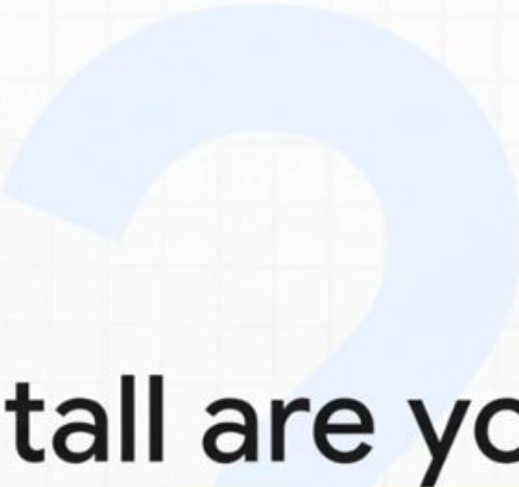
# The Tale of Two Rulers

## IMPERIAL



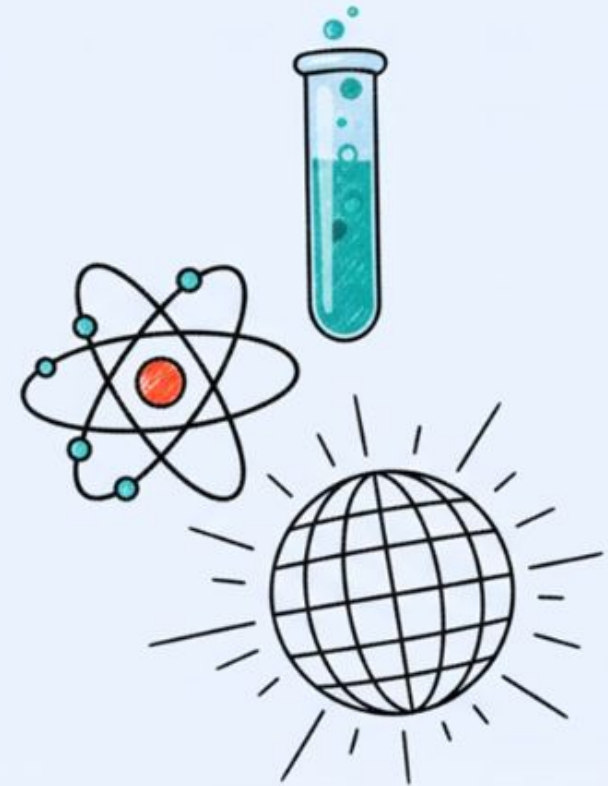
## METRIC





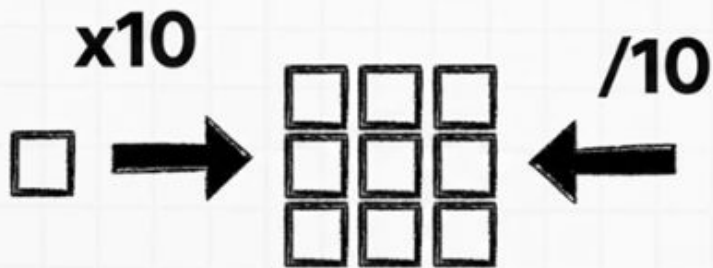
How tall are you? In  
**feet** or in **centimeters**?







# 10





| Prefix     | Meaning | Relation to Meter |
|------------|---------|-------------------|
| Kilo- (k)  | 1000    | $10^3$            |
| Centi- (c) | 1/100   | $10^{-2}$         |
| Milli- (m) | 1/1000  | $10^{-3}$         |

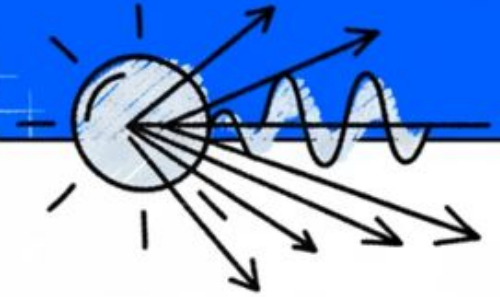






# The Meter's Journey

A Universal Ruler



# Evolution of the Meter



1790s

Defined as 1/10,000,000 of the distance from the North Pole to the Equator.

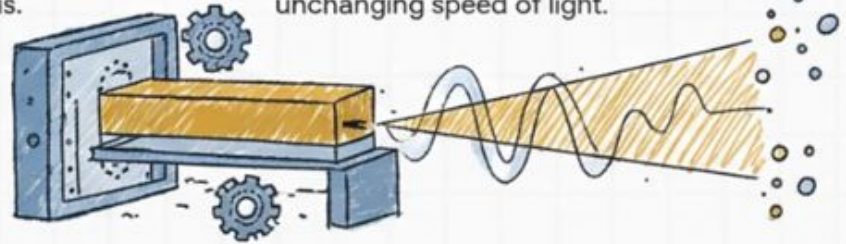


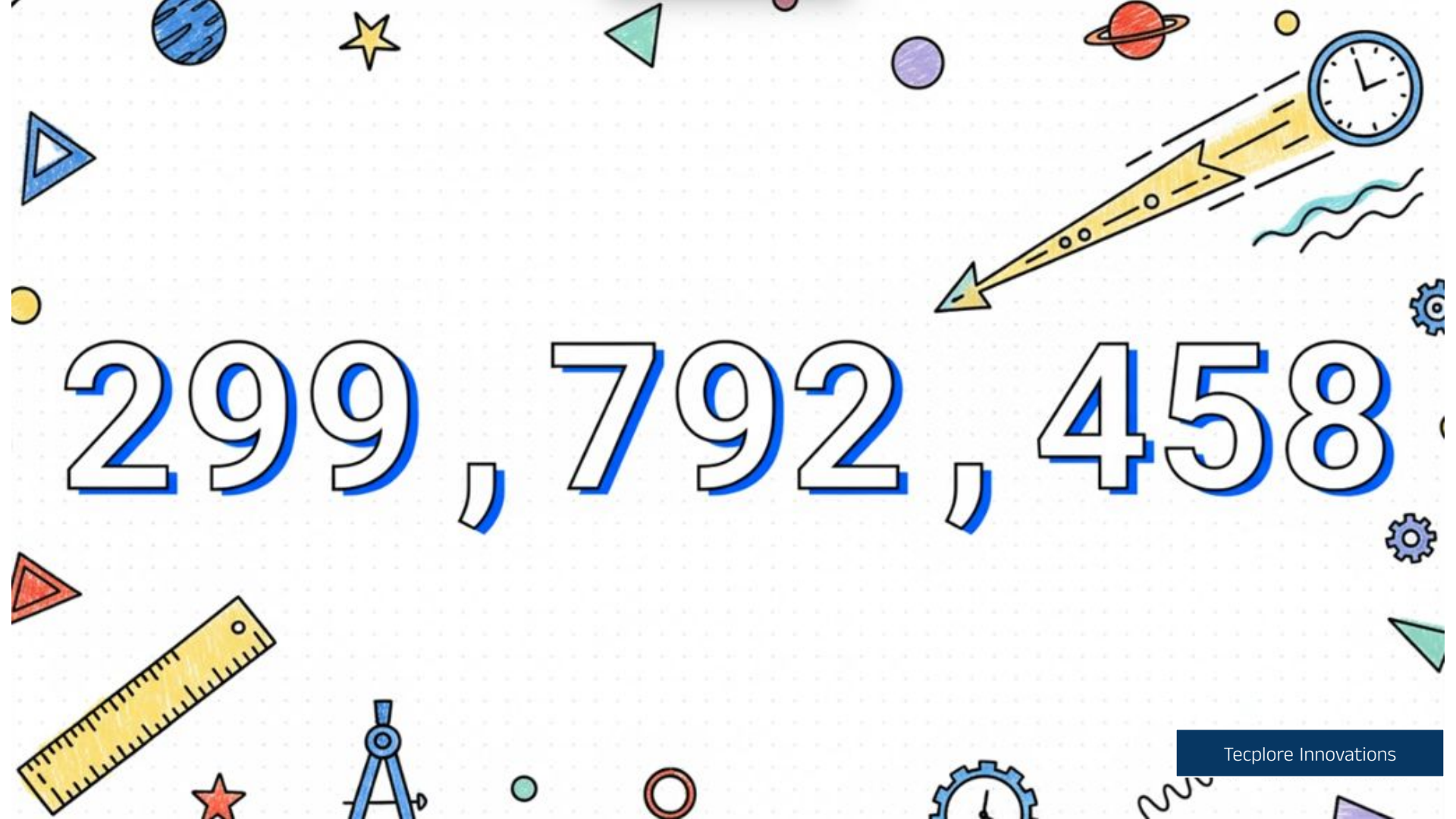
1889

The physical length of a Platinum-Iridium bar kept in a vault in Paris.

1983

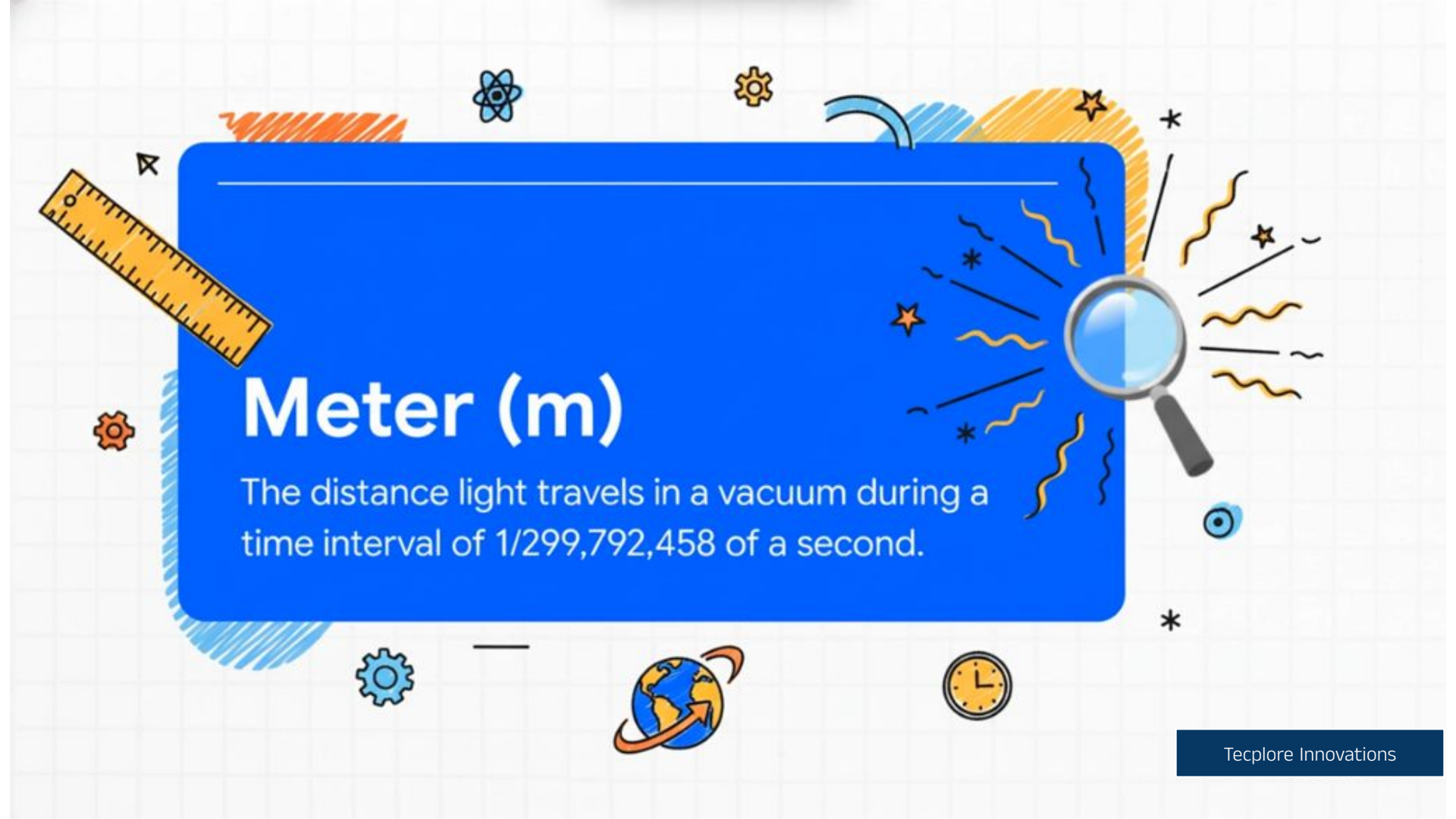
Defined by a universal constant: the unchanging speed of light.





299,792,458





# Meter (m)

The distance light travels in a vacuum during a time interval of  $1/299,792,458$  of a second.

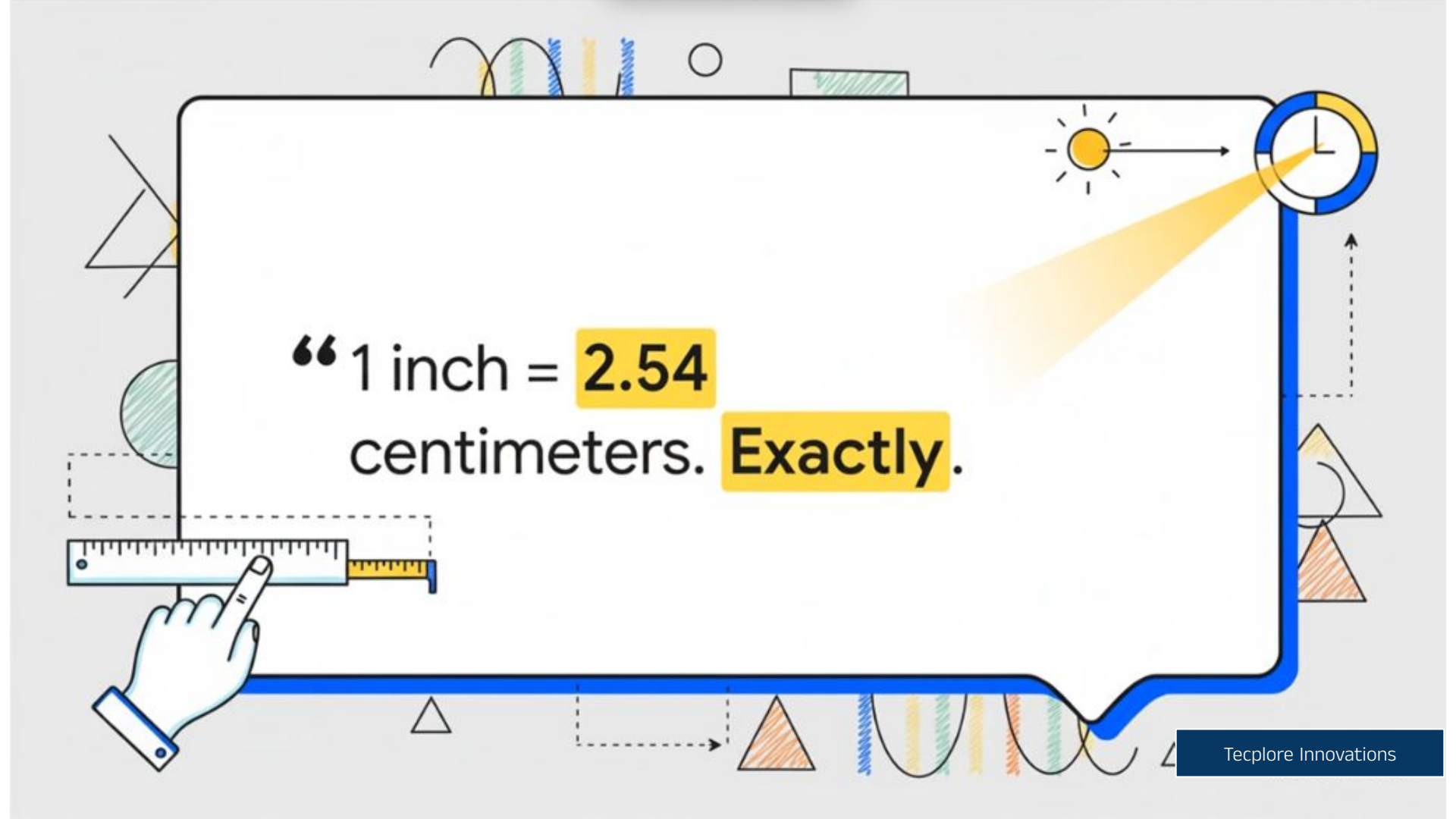
# 2



## The Conversion Bridge

Connecting Two Worlds





“1 inch = 2.54  
centimeters. Exactly.

# Feet to Meters

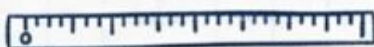


## 1. Feet to Inches

Multiply by 12 (e.g., 5 ft \* 12 = 60 in)



12x



## 2. Inches to Cm

Multiply by 2.54 (e.g., 60 in \* 2.54 = 152.4 cm)



2.54x

/100

## 3. Cm to Meters

Divide by 100 (e.g., 152.4 cm / 100 = 1.524 m)



af







# 3

## Why It Matters

The Takeaway

